

Kevin Le

Systems Engineer

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EDUCATION

University of California, Davis

Davis, CA

Master of Science in Computer Engineering | GPA: 4.0

Sep 2024 – Dec 2026

University of California, Davis

Davis, CA

Bachelor of Science in Computer Engineering | GPA: 3.75

Sep 2021 – Jun 2025

- Deans Honor's List — Tau Beta Pi Scholar

RELEVANT SKILLS

Languages: C, C++ (C++20), Python, Rust, Verilog, RISC-V, Javascript

Developer Tools: Git, CMake, gdb, Kubernetes, Docker, AWS, Nsight Compute (NCU)

Miscellaneous: CUDA, Linux, ROCm, PyTorch, Triton, OpenMPI, OpenMP, OpenCL, FPGA

WORK EXPERIENCE

AggieWorks: RoomU Team

Sep 2023 – Dec 2024

Software Engineering Intern

- Updated a Typescript code base using a React framework to implement critical messaging features for a social network app aimed to connect students looking for roommates
- Worked within a small cross-functional team of developers and designers to reform the application's features and improve user retention
- Interfaced with an SQL database to support page-caching protocols that reduced page access times nearly 65% in order to improve user experience

UC Davis College of Engineering

Sep 2024 – Mar 2025

Lab Teaching Assistant

- Responsible for leading lab sessions for two separate morning section labs for the Circuits course in the Electrical Engineering department
- Taught proper oscilloscope operation, breadboard best practices, and conducted lab demos to explain lecture materials

PROJECTS

Neural Network in CUDA

- Engineered a custom neural network in CUDA C++, manually implementing forward and backward propagation kernels to replicate PyTorch architecture performance
- Profiled kernel execution to diagnose bottlenecks in Streaming Multiprocessor (SM) utilization and memory throughput, identifying high-latency global memory accesses
- Optimized memory patterns to maximize bandwidth efficiency, achieving 98% classification accuracy and training speeds comparable to highly-optimized PyTorch libraries

2D Map Reconstruction

- Synthesized a computer vision solution using a C++ localization algorithm for accurate map generation of a building's floor plan based on camera data
- Trained an object detection model in PyTorch (YOLO), which could identify and label particular landmarks such as doors or tables that the camera comes across during its survey of the floor
- Implemented a Python script that synchronized trajectory data with the detection model's output to create a 2D floor layout with marked landmarks

Distributed Computing using MPI

- Engineered a distributed C application to render the Mandelbrot Set, utilizing OpenMPI for parallel processing across a multi-node cluster
- Achieved a 3x speedup (from 15s to 5s) by refactoring serial algorithms into parallel workloads and optimizing inter-process communication efficiency
- Automated node orchestration using custom Bash scripts to dynamically provision available compute resources and scale processing power on demand